SAILAB BASAK

Data Engineer | Snowflake Development & Administration · Azure Data Factory · Data Quality Engineering

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PROFESSIONAL SUMMARY

Data Engineer with 5 years of experience, including 4+ years designing, building, and administering cloud data platforms on Snowflake and Azure. Develops analytical data models and ELT pipelines end to end — dimensional and star-schema design, SQL transformation, data migration, and query performance tuning — while also owning the platform side: RBAC and role hierarchy, masking policies, warehouse rightsizing, and credit cost governance. Complements this with enterprise data quality and reconciliation engineering, and works fluently with AI agents and prompt engineering to translate business requirements into working data logic. Operates comfortably between product engineering, client data teams, and business stakeholders to turn ambiguous requirements into tested, well-documented, production-grade data assets.

CORE SKILLS

Snowflake Development: Dimensional and Star/Snowflake schema design, SQL transformations, views and stored procedures, Streams & Tasks, Time Travel, Zero-Copy Cloning, query and workload performance tuning

Snowflake Administration — FinOps & Security: Role-Based Access Control and role hierarchy design, dynamic data masking policies, user and role provisioning, warehouse rightsizing, resource monitors, credit usage auditing and cloud cost optimization

Data Architecture & Migration: End-to-end data migration — source extraction, staging, transformation, and cutover validation — building high-performance analytical repositories; SCD patterns, relational modeling

Data Integration & Orchestration: Azure Data Factory, ETL/ELT pipeline design, REST & GraphQL API integration and testing (Postman)

Data Quality Engineering: iceDQ — Validation, Duplicate, Reconciliation, Checksum, Pushdown & Script rules; workflows, schedules, exception reporting; reconciliation frameworks, data profiling, anomaly detection

AI Agents & Prompt Engineering: Fluent day-to-day use of AI agents— structuring and refining prompts to translate business requirements into working SQL, data quality rules, and automation; LLM-assisted development, code review, test-case generation, and documentation

Programming: SQL (advanced), Python (basic)

Tools & Methods: Git, Postman, Power BI, Agile, technical documentation, stakeholder communication

PROFESSIONAL EXPERIENCE

Data Engineer — iceDQ.ai **Sept 2025 – Present**

Client engagement: John Wiley & Sons — global publishing and research

- Design and build **Snowflake environments end to end** for proof-of-concept and demonstration workloads — modeling representative datasets, loading and transforming them, and layering end-to-end data quality validation on top.
- Administer the organization's Snowflake account — **role hierarchy and RBAC design**, dynamic data masking policies, user and role provisioning, warehouse rightsizing, resource monitors, and credit usage auditing for cost optimization across POC, demo, and development accounts.
- Wrote and optimized **complex SQL** to prepare, transform, and cleanse datasets for validation, and performed data profiling and root-cause analysis to isolate discrepancies and drive fixes with upstream data teams.
- Implemented an automated **ingestion-to-validation mechanism** — OAuth-authenticated source retrieval, CSV expansion, timezone-aware date parsing, and dynamic parameter injection into scheduled validation rules — replacing a manual download-and-reformat process.
- Perform **API testing of the iceDQ platform using Postman** — validating REST endpoint behaviour, authentication flows, request/response payloads, and error handling across releases.
- Use **AI agents and structured prompt engineering** to take business requirements through to implementation — generating and reviewing SQL, designing test cases, troubleshooting pipelines, and producing documentation — then validating every output against live data before it ships.
- Partnered with client data teams to translate business rules into concrete validation, exception, and reconciliation logic, and authored migration documentation for promoting rule sets from lower environments into production.

Data Engineer — Infosys Ltd. | Client: Talbots **Mar 2022 – Sept 2025**

Retail analytics — sales, inventory, and customer journey data

- Designed, developed, and optimized **Snowflake data warehouses** — databases, schemas, fact and dimension tables, views, and stored procedures — supporting large-scale enterprise analytics across sales, inventory, and customer journeys.
- Managed **end-to-end data migration** from heterogeneous source systems into Snowflake — extraction, staging, transformation, and cutover validation — building high-performance analytical repositories for downstream reporting.
- Built and maintained production **ETL/ELT pipelines in Azure Data Factory** to ingest data from heterogeneous source systems into Snowflake, orchestrating end-to-end workflows for scheduled loading and transformation.

- Wrote and tuned complex SQL, applying Snowflake features such as **Time Travel, Zero-Copy Cloning, Streams, and Tasks** to improve performance and reduce compute consumption on high-volume workloads.
- Performed **dimensional and relational data modeling** to design scalable structures aligned to business reporting and analytics requirements.
- Implemented **role-based access control and data governance policies** within Snowflake to enforce data security, compliance, and controlled access across teams.
- Managed SQL changes across environments using **Git**, participating in peer code reviews to ensure consistent, reliable releases; authored and maintained schema and data model documentation that accelerated team onboarding.

CERTIFICATIONS

- **SnowPro Core Certification (COF-C02)** — Feb 2025 · **Microsoft Certified: Azure Fundamentals (AZ-900)** — Aug 2024
- Infosys Certified: Microsoft Power BI Developer (Jun 2024) · Data Management Associate (Aug 2024) · Business Intelligence Associate (Jul 2024) · MySQL Associate (Feb 2024)

PROFESSIONAL DEVELOPMENT

- **Master LLM Engineering & AI Agents** — ongoing structured coursework on LLM engineering and agentic workflows, applied to data engineering and data quality automation.

EDUCATION

Sikkim Manipal Institute of Technology — Sikkim, India

Master of Computer Applications — Aug 2019 – May 2021 | CGPA 9.1

Bachelor of Computer Applications — Aug 2016 – May 2019 | CGPA 9.3